

```

Restore partition from image file
Partition to restore: /dev/sda1
Size of partition to restore: 20.00 GiB = 2147483648 bytes
Current image file: /mnt/cdrom/image1.tar.gz.000
File system: ext3fs-3.5

Partition was on device: /dev/sda1
Image created on: Wed Aug 12 02:32:20 2009
Size of the original partition: 20.00 GiB = 2147483648 bytes

Time elapsed: 3m0s
Estimated time remaining: 30m0s
Speed: 45.00 MiB/s
Data copied: 3.25 MiB / 32.76 MiB

```

Figure 7: Restore in progress

 **Tip:** When an image is created for a partition holding the MBR, only MBR can be restored to the system from that image at a later time, by choosing the option “Restore an MBR...” from the initial screen (refer to Figure 5).

Some machines may come with a single partition or not enough space on any of the existing partitions. One solution that was mentioned earlier is to use NFS/Samba to store images on a remote machine. Another simple solution is to use the `partimage-server`. A remote machine running `partimage-server` can be used to store images whose IP address and port can be specified during backup/restore of a partition.

Install `partimage-server` on any distribution and run it using the command `partimaged`. By default, images are stored in `/var/lib/partimaged/`. You can use the `-d` or `--dest` options to specify any other directory to store images on the server. Similarly, you can change the default port number 4025 with the `-p` or `--port` option.

In order to connect to `partimage-server` during backup/restore operations, simply opt for ‘Connect to Server’ in the initial screen (refer to Figure 5) and enter the IP address of the server and the encryption mode. Also change the port number to connect to if the server is using a port other than the default one.

System Rescue CD

A good utility that combines many of the tools discussed here in one CD is the System Rescue CD distro. It comes with many additional utilities like:

- An SSH server to allow other systems to log in remotely and copy files
- Filesystem checkers, scanners, mount and view contents of other partitions
- `mkfs` to create a new filesystem and `sfdisk` to set up partitions
- `parted` and `GParted` for partition manipulation
- Firefox/Dillo for instant browsing
- PXE server capability

Tips and tricks

1. To set up a network manually on any live CD, use the following code:

```

Starting network scripts...
***** SystemRescue-CD ***** 1.6.0 ***** ttg@rk **
http://www.sysresccd.org

Type net-setup eth0 to specify ethernet configuration.
If you have a corporate ethernet network, you can configure by hand:
ifconfig eth0 192.168.x.x (your static IP address)
route add default gw 192.168.x.x (IP address of the gateway)
To start an ssh server on this system, type /etc/init.d/sshd start.
You will need to create an user or to change the root password with passwd.
Available console editors: nano, vim, emacs, joe
Graphical web browser in the console: links -g www.web-site.org.
WARNING: Never mount anything on /mnt! It could freeze the system.
Use fdisk /mnt/majordisk and mount on /mnt/majordisk instead.
Mkfs -3y: If you need a full Read-Write NTFS access, use Mkfs-3g.
Format the disk: mkfs-3g /dev/sda1 /mnt/majordisk
Graphical consolemount: I use either Xorg or Xvesa.
Type utdmd to run the graphical environment (for starts but it may fail)
X-3g comes with Window-Manager and you can use several graphical tools:
Partition manager: gparted
Web browser: ..... Firefox-0.9 and dillo
Text editors: ..... gvim and leafpad
02:50 root@sysresccd: root #

```

Figure 8: System Rescue CD

ifconfig eth0 up

Use a different device number other than `eth0`, if required. On the System Rescue CD you can use the `net-setup eth0` command instead.

2. While running `g4u`, make sure of your disk IDs by running the `parts` command first. For example, if the original disk is detected as `wd1`, and the second one as `wd0`, running `copydisk wd0 wd1` will mean losing all your data in the original disk. Or a safer method is erasing all partitions in the target disk before cloning, by using `GParted` or any similar utility. Now `parts` on the target disk will not show any partitions, whereas on the original disk, partitions will be listed.
3. In the System Rescue CD, if your `partimage` is already running, go to another virtual console (using `Ctrl+Alt+F2`, `Ctrl+Alt+F3`, etc) for mounting, checking image paths, etc. Use `Ctrl+Alt+F1` to come back to the first console.
4. Don't try to save an image in the same partition you're backing up.
5. As Linux distros are compatible with many configurations, cloned disks or partitions work fine in systems with different configurations also — with the exception of one or two features. For example, an image taken for a Linux distro on a Core 2 Duo system works well with PIII or AMD machines also, only requiring you to tweak `/etc/X11/xorg.conf` according to new display settings, if you have a problem with starting the GUI. 

References

- GParted: GParted.sourceforge.net
- g4u: www.feyrer.de/g4u
- Partimage: partimage.org
- System Rescue CD: www.sysresccd.org

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